



JET SOCIAL & COMMUNITY IMPACT

Decarbonization & Livelihoods Matrix

Presented by: **Sipho Wiseman Khuzwayo**, Founder & Managing Director



Part 1: Socioeconomic Revitalization

Empowering vulnerable communities within the Secunda and Kinross agricultural corridors



Tri-Pillar Just Transition Objectives



Labor Migration

Absorbs retrenched regional coal miners, training and reskilling them in clean, high-precision food technologies.



Food Sovereignty

Establishes stable local protein channels, isolating food distribution lines from volatile global supply chains.



Resource Equity

Utilizes off-grid solar and recycled bio-slurry utilities, protecting regional resources from municipal grid strain.



Empowering Vulnerable Demographics

67%

Women-Led Teams

Inclusive Regional Job Growth

In close alignment with DMRE and G20 Principles for Just Transitions, our hiring framework actively corrects historical labor imbalances:

- ✓ **Youth Activation:** 86% of short-term building opportunities are directly awarded to local youth.
- ✓ **Cooperative Synergy:** Off-take pipelines directly integrate with women-led municipal farm co-ops.



Skills Portfolio & Training Academy

1. Green Technology Skills

Accredited technical workshops in solar microgrid maintenance, LFP storage metrics, and anaerobic digester operations.

2. Applied Agritechnology

Hands-on training in biosecurity pod systems, conveyor mechanics, and water sanitization protocols for regional youth.



Securing Regional Protein Supplies

Bridging Mining Decline Gaps

As regional mining operations inevitably wind down, local communities face extreme socioeconomic disruptions and inflation traps.

- ✓ **Stable Output:** Delivers 34,560 broilers and 480 market hogs annually.
- ✓ **Independent Cost Structure:** Off-grid utility designs lock in stable wholesale pricing.
- ✓ **Local Integration:** Direct distribution through regional NPOs and the Mam Dube foundations.





Part 2: Environmental Stewardship

Decarbonization, resource circularity, and biological waste diversion



Decarbonization & Carbon Savings

Traditional Grid Farm (Coal)

12.4 Tons CO₂e / Year (100%)

LWP Bio-Solar closed loop

1.5 Tons
CO₂e /
Year
(12.1%)

By replacing coal-grid electrical heating with recycled sub-floor steam heat, Lwandile slashes emissions by an incredible 87%.



Ecosystem & Resource Performance

Resource Indicator	Standard Baseline Practice	Lwandile Closed Loop Value	Environmental Impact Outcome
Biological Waste	Raw lagoon dumping	Anaerobic biogas parsing	Prevents local groundwater contamination risks.
Organic Fertilizer	Chemical import purchases	Dry bio-slurry packing	Restores regional soil health and carbon storage.
Hydration Delivery	Continuous grid pumping	Gravity water tower storage	Saves more than 40,000 liters of active pumping energy.



Regional Impact & Scaling Path





Executive Visionary Voice

“

We are showing South Africa that a Just Energy Transition is not merely about decommissioning power stations; it is about building the next generation of food, energy, and community sovereignty.

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— Sipho Wiseman Khuzwayo, Founder & MD



ASITHOKOZE- LET US EXECUTE

Sipho Wiseman Khuzwayo | Founder & MD

Precision Alliance Development Pipeline | Just Energy Transition